

**"THE GUARDIAN": A MODULAR AI FOR
PROACTIVE HUMAN-LAYER CYBERSECURITY ON
SOCIAL MEDIA.**

Project ID: 25-26J-041

Project Proposal Report

De Alwis, H. P. G. H. S.

B. Sc. (Hons) Degree in Information Technology (Cyber Security)

Department of Information Technology
Sri Lanka Institute of Information Technology
Sri Lanka

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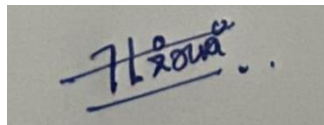
Sri Lanka Institute of Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Date:

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30/08/2025

Abstract

Misinformation on social media has become a significant human-layer cybersecurity threat, exploiting cognitive biases, limited media literacy, and uneven awareness across user populations. Existing detection systems typically rely on fact-checking databases or binary classification outputs, offering little personalization or guidance to the end user. This project proposes a misinformation detection module as part of *The Guardian*—a modular, AI-powered browser extension designed to provide real-time human-layer security. The module employs transformer-based NLP models, such as BERT and DistilBERT, fine-tuned on misinformation datasets including LIAR and CoAID, to detect manipulative or misleading text highlighted by the user. Unlike conventional tools, this module incorporates user profiling based on metadata such as education level, prior awareness of misinformation, and domain knowledge. When misinformation is detected, the system initiates a chat interface that adapts its explanation style and level of detail to the user’s profile. By focusing on guidance rather than censorship, the module promotes critical thinking and media literacy, aiming to strengthen user resilience against misinformation in a personalized and scalable manner.

Keywords: Cybersecurity, Natural Language Processing, Human-Layer Security, Misinformation, User Profiling.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
B2B	Business to Business
B2C	Business to Consumer
BERT	Bidirectional Encoder Representations from Transformers
CoAID	COVID-19 Healthcare Misinformation Dataset
CSI	Capture, Score, Integrate (from CSI Fake News Model)
DBIR	Data Breach Investigations Report
EMNLP	Empirical Methods in Natural Language Processing
FEVER	Fact Extraction and Verification
FR	Functional Requirement
F1-Score	Harmonic Mean of Precision and Recall
ICWSM	International AAAI Conference on Web and Social Media
JSON	JavaScript Object Notation
LIAR	Liar Dataset for Fake News Detection
NFR	Non-Functional Requirement
NLP	Natural Language Processing
PII	Personally Identifiable Information
OpSec	Operational Security
PyTorch	Python Torch (Deep Learning Framework)
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
UX	User Experience
WBS	Work Breakdown Structure

1. INTRODUCTION

The focus of cybersecurity has increasingly shifted from protecting infrastructure to addressing the vulnerabilities of the human user, who remains the most unpredictable element of the security ecosystem. Among the most pervasive of these human-layer threats is misinformation, which exploits cognitive biases, limited awareness, and uneven access to reliable information. Social media platforms, while invaluable for connectivity and information exchange, have simultaneously become fertile grounds for large-scale misinformation campaigns that erode trust in institutions, distort public opinion, and in some cases, endanger public health and national security [1]. Unlike traditional cyber threats that target systems, misinformation manipulates human cognition directly, making it uniquely challenging to mitigate.

The scale of the challenge is amplified by psychological phenomena such as confirmation bias, where individuals are predisposed to accept information consistent with their beliefs, and by the tendency of emotionally manipulative content to spread more rapidly than factual corrections [2]. High-profile cases, such as the widespread circulation of false claims during the COVID-19 pandemic, demonstrate the devastating consequences of unchecked misinformation [3]. Existing platform-based moderation tools remain largely reactive, relying on centralized fact-checking databases or crowdsourced reporting, which are insufficient for real-time, user-centric protection [4].

A critical research gap exists in the development of misinformation detection tools that go beyond binary outputs and adapt explanations to the characteristics of the individual user. Current systems typically treat all users as homogeneous, ignoring factors such as educational background, prior media literacy, and domain-specific knowledge, which influence how misinformation is perceived and understood. This project proposes a misinformation detection module within The Guardian, a modular browser extension for proactive, client-side defense against human-layer threats. Leveraging transformer-based NLP models fine-tuned on misinformation datasets, the module identifies manipulative or misleading content in real time. Its novelty lies in its integration of user profiling based on metadata, enabling explanations that are

contextually adapted to the user's knowledge level. Rather than censoring information, the module delivers personalized guidance through a chat interface, functioning as a cognitive “nudge” to enhance awareness and critical evaluation of online content.

2. BACKGROUND & LITERATURE SURVEY

This section critically reviews the existing body of research related to misinformation detection on social media. It evaluates the role of human factors, examines automated detection techniques, outlines the methodological foundations for misinformation analysis, and identifies the research gap addressed by this project, namely the need for user-profile-aware, personalized explanations.

2.1 The Human Factor and Social Media as a Threat Vector

Cybersecurity in the digital age increasingly revolves around human vulnerabilities rather than purely technical exploits. Ghafir et al. [1] highlights the "human factor" as a central weakness, with social media providing a uniquely fertile environment for manipulation. Platforms are deliberately designed to encourage rapid sharing and engagement, amplifying emotionally charged content that frequently includes misinformation [2]. Unlike conventional cyberattacks that target systems, misinformation exploits psychology and cognitive biases, manipulating perception and behavior directly [3].

Research demonstrates that misinformation thrives on psychological mechanisms such as confirmation bias, selective exposure, and motivated reasoning [4]. These predispositions not only increase users' likelihood of believing false content but also reduce the effectiveness of corrective interventions. Compounding this issue, misinformation spreads more quickly and broadly than factual corrections, especially when framed with emotionally manipulative or sensational language [5]. This makes misinformation one of the most persistent human-layer threats in cyberspace, with impacts ranging from public health crises to political destabilization.

2.2 Automated Detection of Misinformation

Automated misinformation detection has been approached from multiple perspectives, with fact-checking systems and machine learning models dominating the field. Knowledge-driven approaches rely on curated databases of verified claims (e.g., PolitiFact, Snopes), but these are inherently reactive and cannot scale to the dynamic, evolving nature of misinformation [6]. To overcome this limitation,

machine learning and NLP-based methods have been widely adopted. Wang’s LIAR dataset [7], containing labeled political claims, and the COVID-19 focused CoAID dataset [8], have been instrumental in training classifiers capable of distinguishing true from false information.

Transformer-based architectures, including BERT and RoBERTa, have emerged as state-of-the-art solutions for misinformation detection due to their ability to capture contextual and semantic subtleties [9]. However, these systems face challenges in real-world deployment: sarcasm, coded language, and novel misinformation narratives often reduce model accuracy [10]. More critically, most existing tools provide binary outputs (“true/false”), with little attention to how feedback is communicated to end users. This limits their effectiveness in practice, as users are not guided toward better judgment but are merely presented with categorical results.

2.3 Core Technologies and Methodological Foundations

The methodological foundations for misinformation detection lie in natural language processing (NLP) and deep learning. Transformer-based models such as BERT [9] have significantly advanced the ability to analyze manipulative linguistic cues by learning contextual representations. DistilBERT, a lighter alternative, enables real-time classification suitable for browser-based applications. Benchmark datasets such as LIAR [7], CoAID [8], and FakeNewsNet [11] provide labeled misinformation examples across political, health, and general domains, supporting fine-tuning and comparative evaluation of these models.

While linguistic analysis forms the technical core, user profiling introduces an additional dimension. Prior research in media literacy and misinformation resilience shows that educational background, domain knowledge, and prior exposure to misinformation significantly affect user vulnerability [12]. Metadata such as self-reported knowledge level, browsing behavior, or demographic indicators can therefore inform how systems deliver feedback. Incorporating such profiling into misinformation detection systems bridges the gap between purely technical detection and effective user guidance, laying the groundwork for personalized interventions.

2.4 The Profiling Gap and the Need for User-Centric Design

Although significant research has been devoted to improving model accuracy in misinformation detection, a critical gap remains in user experience and personalization. Most existing systems treat users as homogeneous, providing one-size-fits-all warnings or classifications. This ignores the diversity of users in terms of education level, prior awareness, and ability to critically evaluate information [13]. Evidence from media literacy studies indicates that tailoring interventions to user knowledge levels can significantly increase their effectiveness [14].

This project directly addresses this gap by integrating metadata-based user profiling into the misinformation detection module. Instead of delivering uniform binary outputs, the module adapts explanations through a conversational chatbot interface, aligning the complexity and depth of feedback with the user's profile. For instance, a user with low awareness may receive detailed, example-based explanations, while a more knowledgeable user may be given concise confirmations and links to reliable sources. This approach shifts the focus from censorship to education, providing a personalized cognitive "nudge" that strengthens long-term resilience against misinformation without overwhelming or alienating users [15].

3. RESEARCH GAP

The literature survey highlights several critical shortcomings in existing research and practical tools for combating misinformation on social media. While progress has been made in automated detection and platform-level moderation, the following distinct gaps remain:

- **Over-Reliance on Fact-Checking Databases:** Most existing misinformation detection systems depend on centralized fact-checking repositories (e.g., PolitiFact, Snopes) [6], which are reactive, limited in scope, and unable to keep pace with the dynamic and rapidly evolving nature of misinformation. There is a notable lack of approaches that operate independently of fact-checking databases and can identify manipulative or misleading language patterns in real time.
- **Neglect of User Profiling in Explanations:** Current research predominantly focuses on improving model accuracy (e.g., through Transformer-based architectures [9]) but treats users as homogeneous. Little attention has been given to tailoring interventions according to users' knowledge levels, media literacy, or educational background. This creates a gap in systems capable of adapting feedback to the individual user's profile, thereby enhancing comprehension and trust.
- **Binary and Non-Explanatory Outputs:** Many state-of-the-art misinformation classifiers provide only categorical judgments ("true" or "false"), without offering explanatory insights into *why* a text may be misleading [7], [10]. This reduces transparency and fails to encourage critical thinking, leaving users reliant on the system's authority rather than learning to evaluate content themselves.
- **Lack of Integrated, User-Centric Educational Tools:** Existing solutions are fragmented, either limited to platform-level moderation or academic prototypes that address a single detection task. There is a clear need for an integrated, user-facing module that not only detects misinformation but also

provides personalized, real-time explanations, functioning as an educational tool rather than a censorship mechanism.

- **Gap in Usability and Trust-Centered Design:** Research has established that overwhelming or intrusive warnings can cause “security fatigue” [12], reducing user compliance. Yet, few misinformation detection tools explicitly integrate principles of usable security into their design. A gap exists for solutions that adopt subtle, conversational interfaces such as chat-based explanations that deliver guidance without disrupting the user experience.

4. RESEARCH PROBLEM

Based on the identified research gaps, this project addresses a central research problem that can be articulated as follows:

How can transformer-based Natural Language Processing (NLP) models, fine-tuned on misinformation datasets and integrated with user profiling, be designed and implemented to provide real-time, proactive, and educational feedback on misinformation within the dynamic and informal linguistic environment of social media?

This overarching problem encompasses several sub-problems that this research will investigate:

1. **How to adapt and fine-tune pre-trained Transformer models** (e.g., BERT, DistilBERT) to detect misinformation expressed in informal, noisy, and context-dependent social media text?
2. **How to incorporate metadata-based user profiling** (e.g., educational background, media literacy, domain knowledge) to tailor the depth and style of explanations for different users?
3. **How to design an explanation mechanism** that moves beyond binary classifications by highlighting manipulative linguistic features (e.g., exaggeration, emotional bias) to promote user understanding and critical thinking?
4. **How to integrate detection and explanation into a browser-based module** that operates in near real-time and presents feedback through a non-intrusive, chat-based interface?
5. **How to evaluate the effectiveness of personalized, profile-aware explanations** in improving user awareness and trust, while minimizing the risks of information overload or “warning fatigue”?

5. OBJECTIVES

This research aims to address the identified research problem through the achievement of one main objective, which is further broken down into four specific, measurable sub-objectives.

5.1 Main Objective

To design, implement, and evaluate a robust, real-time misinformation detection module for The Guardian system that accurately identifies manipulative or misleading content on social media and provides personalized, educational explanations through a chat-based interface, adapted to user profiles.

5.2 Specific Objectives

To develop and fine-tune a transformer-based misinformation detection model by [End of Month 4]. This will be achieved by leveraging pre-trained NLP models (e.g., BERT, DistilBERT) and fine-tuning them on misinformation datasets such as LIAR, CoAID, and FakeNewsNet. The model's success will be measured by achieving an F1-score of at least **0.85** on a benchmark test dataset of annotated social media posts.

To design and implement a user profiling mechanism by [End of Month 5]. This involves capturing metadata such as educational background, prior knowledge of misinformation, and media literacy level to create user profiles. The profiling engine will adapt explanation styles (e.g., detailed, example-based, or concise) according to these profiles. The success of this objective will be measured by the system's ability to consistently generate user-appropriate responses in evaluation scenarios.

To integrate the misinformation detection and user profiling components into a unified, efficient API endpoint by [End of Month 6]. This will be accomplished by wrapping the detection and profiling engines into a Python-based service and exposing their functionality via a FastAPI endpoint. The endpoint must accept highlighted text as input and return a structured JSON object containing detection results and personalized explanation guidance. Performance will be measured by

ensuring that the average processing and response time for a typical social media post (under 280 characters) is less than 500 milliseconds.

To conduct a comprehensive performance and usability evaluation of the completed module by [End of Month 7]. This involves a quantitative analysis of the model's accuracy, precision, recall, and F1-score, alongside a qualitative user study assessing the clarity, trustworthiness, and usefulness of the chatbot's explanations. The final evaluation will include a comparative analysis of model architectures (e.g., BERT vs. DistilBERT) and user feedback on the effectiveness of personalized explanations in improving awareness.

6. METHODOLOGY

This section outlines the systematic approach that will be followed to achieve the project's objectives. It details the overall system architecture of The Guardian, the specific internal workflow of the Misinformation Detection Module, the data collection and preparation strategy, and the plan for rigorous evaluation.

6.1 Overall System Architecture

The proposed *Guardian* system is designed as a client-server application, comprising a lightweight browser extension (the client) and a powerful backend analysis engine (the server). This architecture is chosen to ensure real-time responsiveness while handling computationally intensive natural language processing tasks securely on the backend. The workflow, as depicted in Figure 1, is as follows:

1. **Text Highlighting & Extraction:** The browser extension allows the user to highlight suspicious social media text and forwards the selection for analysis.
2. **Secure API Request:** The highlighted text is transmitted via HTTPS to the backend API. Only the selected text and an anonymous user identifier are sent; no personal metadata is stored.
3. **Backend Processing:** The FastAPI backend receives the request and passes the text to the four specialized analysis modules.
4. **Modular Analysis:** Each module (PII/OpSec, Hate Speech, Misinformation, Scam) processes the text and returns a structured JSON response indicating any detected threats. (The Misinformation Module processes the text, detecting manipulative techniques and possible misinformation.)
5. **Aggregated Response:** The backend combines results from all modules into a single JSON object and sends it back to the browser extension.
6. **User Notification:** T The extension displays a subtle warning icon next to the highlighted text. Clicking the icon opens a chat-based explanation window that provides the user with an educational response.

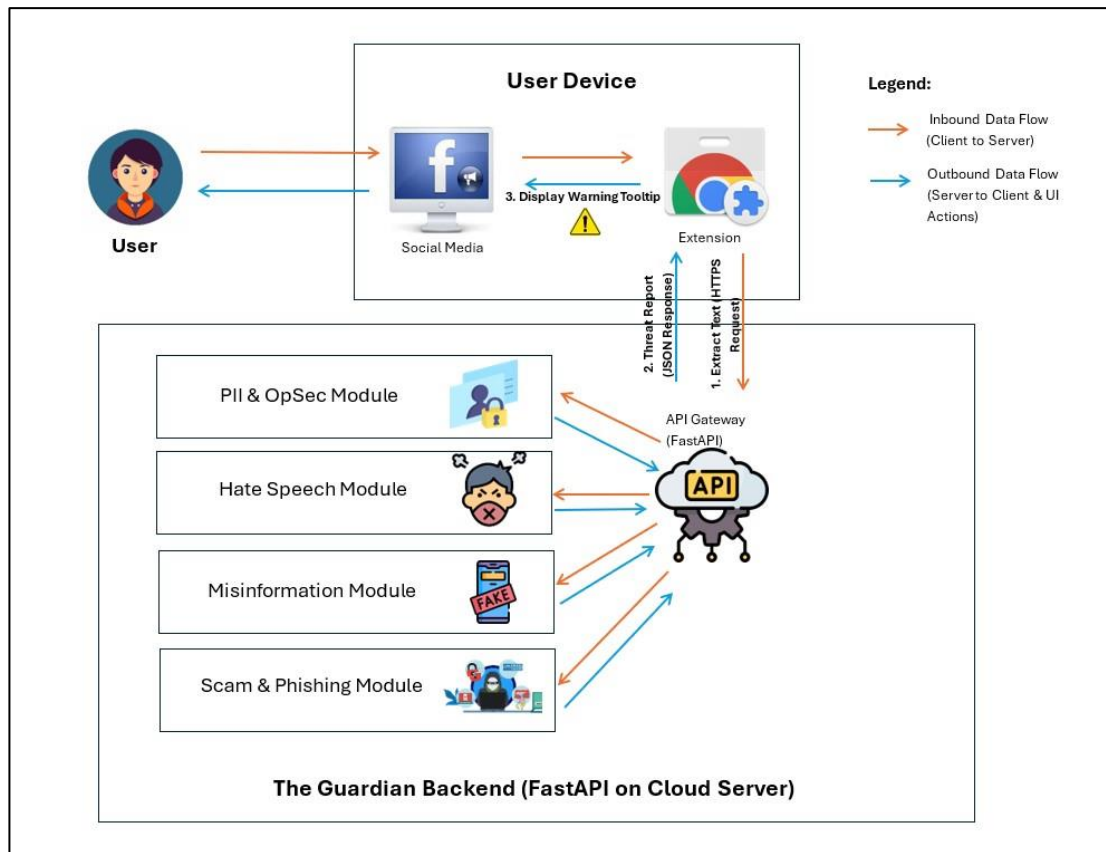


Figure 6.1: High-level system architecture of "The Guardian," showing the data flow from the client-side browser extension to the backend analysis modules and back.

6.2 Misinformation Module: Design and Implementation

The Misinformation Module is the core component of this research and will be developed using a hybrid approach to maximize accuracy, explainability, and usability. Its internal workflow, illustrated in Figure 2, is designed as a multi-stage pipeline

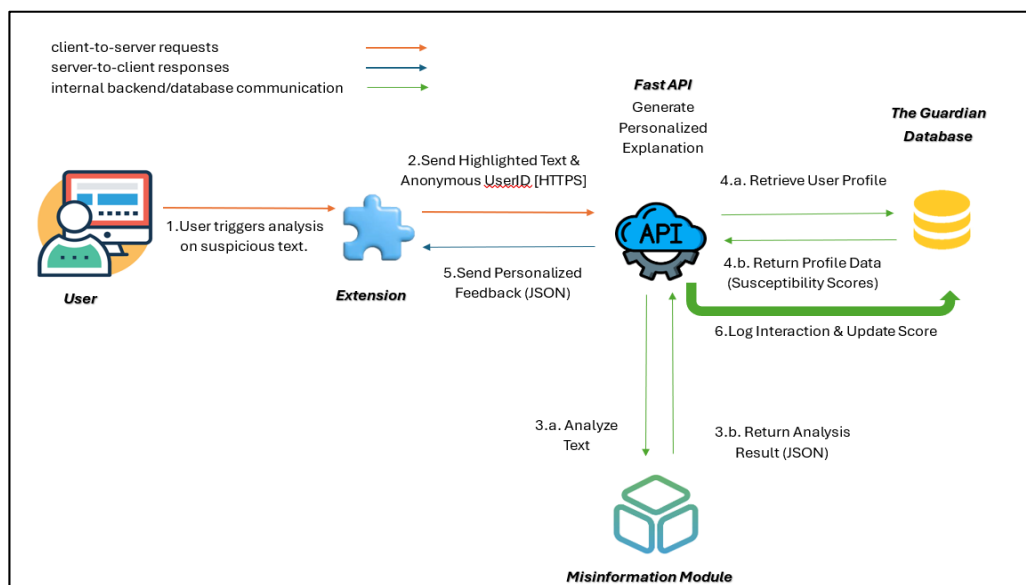


Figure 6.2: High-Level System Flow of the Misinformation Detection Module in The Guardian

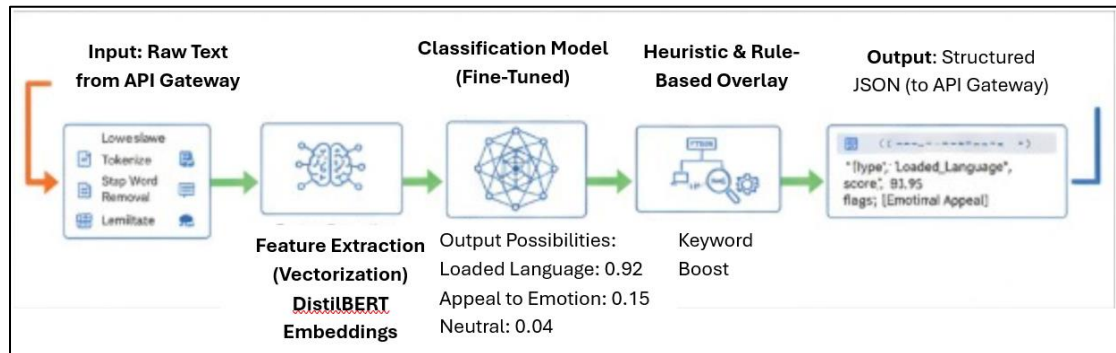


Figure 6.3: Expanded View of Inside the Misinformation Detection Module

Stage 1: Misinformation Detection (Transformer-based NLP)

- **Model Selection:** A pre-trained Transformer model (e.g., DistilBERT) will be fine-tuned for misinformation classification due to its balance of efficiency and accuracy
- **Classification Task:** Instead of binary “true/false” claims, the model will perform multi-label classification to detect manipulative techniques such as Loaded Language, Appeal to Emotion, Exaggeration, or Conspiracy Framing.
- **Confidence Score:** The model outputs not only classification labels but also confidence scores for each technique.

Stage 2: Heuristic Overlay (Rule-Based Analysis)

- **Keyword Dictionaries:** Rule-based heuristics will supplement the model by detecting common misinformation markers (e.g., “secretly planning,” “miracle cure,” “guaranteed proof”).
- **Precision Boost:** This layer increases confidence in predictions and helps reduce false positives.

6.3 Data Collection and Preparation

A domain-specific dataset will be created by combining publicly available misinformation datasets with synthetic examples.

1. **Corpus Acquisition:** Benchmark misinformation datasets such as LIAR, FakeNewsNet, and CoAID will form the base corpus.
2. **Data Annotation:** A subset will be re-annotated to capture manipulation techniques beyond simple true/false labels.
3. **Dataset Partitioning:** The dataset will be split into Training (70%), Validation (15%), and Test (15%) sets for robust evaluation.

6.4 Technical Requirements & Resources

The project will leverage open-source software and standard computing resources. No specialized hardware or budget is required.

- **Software & Libraries:** Python (v3.8+), Hugging Face Transformers, PyTorch/TensorFlow, spaCy, and FastAPI for backend development.
- **Development Environment:** Personal laptops and university lab machines with at least 16GB RAM. Version control is managed via GitHub.
- **Data Sources:** Public misinformation datasets (e.g., LIAR, CoAID, FakeNewsNet, Kaggle repositories) and synthetic examples created for testing.

6.5 Evaluation Plan

The module will be evaluated using both quantitative and qualitative methods:

- **Misinformation Detection Performance:**
 - Precision, Recall, and F1-Score (target ≥ 0.85).
 - Confidence calibration across multiple misinformation techniques.
- **Performance Benchmarking:**
 - End-to-end latency measured (highlight $\rightarrow$ API $\rightarrow$ JSON $\rightarrow$ UI response).
 - Target: <500 ms for posts under 280 characters.

- **Qualitative Usability Study:**
 - A user study (n=10–15) will be conducted where participants highlight suspicious posts.
 - Data collected via questionnaires and semi-structured interviews.
 - Metrics: clarity of chatbot explanations, trust in system feedback, perceived educational value, and non-intrusiveness of alerts.

7. DESCRIPTION OF PERSONNEL AND FACILITIES

This section details the composition of the project team, the specific roles and responsibilities of each member, and the facilities and resources available for the successful execution of this research.

7.1 Personnel

The project team consists of four final-year undergraduates reading for the B. Sc. (Hons) in Information Technology (Cyber Security). Each member brings a unique specialization in cybersecurity, natural language processing, and software development, enabling a well-rounded approach to project delivery. Roles are clearly distributed to ensure accountability and efficient collaboration.

M1 – Herath, H. M. G. D. K. (IT22361486) – Project Coordinator & PII/OpSec Module Lead

- Oversees project-wide coordination, ensuring milestones are achieved and modules are seamlessly integrated.
- Leads the design and implementation of the PII and OpSec risk detection module, including dataset creation, model training, and evaluation.

M2 – De Alwis, H. P. G. H. S. (IT22317926) – Misinformation Module Lead

- Responsible for investigating linguistic and psychological techniques used in misinformation.

- Develops the misinformation detection module by fine-tuning transformer-based NLP models and integrating user profiling for personalized feedback.

M3 – Fernando, M. R. I. (IT22366054) – Hate Speech Module Lead

- Focuses on identifying hate speech patterns and linguistic markers.
- Designs and implements the AI module specialized in detecting toxic, abusive, or implicitly coded hateful language.

M4 – Vishwa, J. R. (IT22916112) – Scam & Phishing Module Lead & UI/UX Specialist

- Conducts research into scam and phishing strategies prevalent on social media.
- Leads the development of the scam/phishing detection module and oversees the design of the browser extension's interface, including usability testing and wireframing.

7.2 Facilities and Resources

The project will make use of standard academic and open-source resources, ensuring that no costly or specialized infrastructure is required.

- **Laboratory Facilities:** The team will utilize computing laboratories at the Sri Lanka Institute of Information Technology (SLIIT), equipped with the necessary hardware and networking facilities for experimentation and testing.
- **Hardware:** Development and model training will be carried out using a combination of personal laptops and university desktops. These systems meet the necessary requirements for NLP model training and evaluation (e.g., 16 GB RAM).
- **Software:** The entire project will be developed using open-source tools and frameworks:
 - **Programming Language:** Python (v3.8+)

- **Core Libraries:** Hugging Face Transformers, PyTorch/TensorFlow, spaCy, and FastAPI for backend services.
- **Development Tools:** Visual Studio Code for coding, Git and GitHub for version control and collaborative development.
- **Data Sets & Resources:** Misinformation datasets such as LIAR, CoAID, and FakeNewsNet, along with other academic repositories (e.g., Kaggle, Papers with Code), will be employed for training and evaluation. The SLIIT digital library will support literature reviews and access to relevant academic publications.

8. PROJECT REQUIREMENTS

This section defines the specific functional and non-functional requirements for the misinformation detection module. It also identifies the system's intended users and outlines tentative use cases and test cases that will guide the development and validation process.

8.1 User Requirements

The primary user of this module is any individual who uses social media and is concerned about the credibility of the information they encounter, but may lack the time, resources, or technical expertise to critically evaluate every piece of content.

The user requires a tool that:

- **Operates seamlessly with minimal user effort**, allowing text to be highlighted for instant analysis.
- **Provides simple, clear, and educational feedback** on whether highlighted content shows signs of misinformation.
- **Does not censor, block, or store user data**, ensuring that privacy is preserved.
- **Improves the user's awareness and critical evaluation skills** over time by offering personalized, context-aware explanations through a conversational interface.

8.2 Functional Requirements (FR)

Functional requirements specify the concrete actions the system must be able to perform.

Table 8.1: Functional Requirements (FR) for the Misinformation Module

ID	Requirement	Priority
FR1	The system shall allow the user to highlight text within social media posts or comments for analysis in real-time.	High
FR2	The system shall process the highlighted text and classify it as either misinformation or not misinformation using a fine-tuned Transformer-based NLP model.	High
FR3	The system shall identify manipulative linguistic patterns in the text (e.g., exaggeration, emotional triggers, sensational phrasing).	High
FR4	The system shall cross-reference content with relevant misinformation datasets (e.g., LIAR, CoAID, FakeNewsNet) to support classification accuracy.	High
FR5	The system shall generate a structured JSON output containing the detection result, key features (e.g., manipulative terms), and confidence scores.	High
FR6	The system shall provide a visual, non-intrusive indicator on the browser interface next to the highlighted text when misinformation is detected.	Medium
FR7	The system shall, upon user interaction (e.g., clicking the indicator), open a chat-based interface that provides a tailored, educational explanation of why the text may be a misinformation.	High

8.3 Non-Functional Requirements (NFR)

Non-functional requirements define the quality attributes and constraints of the system.

Table 8.2: Non-Functional Requirements (NFR) for the Misinformation Module

ID	Requirement	Metric
----	-------------	--------

NFR1	Performance: The end-to-end misinformation analysis (from highlighted text input to JSON output) for a standard post (<280 characters) shall be completed in under 500 milliseconds .	Average response time
NFR2	Accuracy: The misinformation detection model shall achieve a minimum F1-score of 0.85 on the blind test set.	F1-Score
NFR3	Privacy: The backend service shall not log, retain, or share any user-submitted text post-analysis.	Code review, System design
NFR4	Usability: Alerts and chat-based explanations shall be delivered in a non-intrusive manner that does not obscure social media content or interrupt the browsing experience.	Heuristic evaluation, User feedback
NFR5	Security: All communication between the browser extension and the backend API shall be encrypted using HTTPS .	Network traffic analysis
NFR6	Modularity: The misinformation detection module shall be implemented as an independent, self-contained component with a clearly defined API, enabling standalone testing and seamless integration into The Guardian system.	Code structure review

8.4 Use Cases

Use Case 1: Detection a Misleading Claim

- **Actor:** Social Media User
- **Precondition:** User is browsing social media and highlights a suspicious post or comment.
- **Scenario:**
 1. User highlights the sentence: *"Scientists confirm that drinking garlic water cures COVID-19 instantly."*
 2. The "Guardian" browser extension sends the highlighted text to the backend.

3. The misinformation detection module analyzes the text using a fine-tuned Transformer model (e.g., BERT).
 4. The model classifies the text as misinformation with high confidence and identifies manipulative terms (“cures instantly”).
 5. The module returns a JSON response indicating the detection, confidence score, and key flagged terms.
 6. A subtle warning icon appears next to the highlighted text.
 7. The user clicks the icon, triggering the chat interface. The chatbot explains: *“This statement is flagged as misinformation. No scientific evidence supports garlic water as a cure for COVID-19. The phrase ‘cures instantly’ is often used in manipulative health-related misinformation. ”*
- **Postcondition:** The user is made aware that the highlighted content is misinformation and understands *why* it was flagged.

Use Case 2: Detection of Emotionally Manipulative Content

- **Actor:** Social Media User
- **Precondition:** User is logged into a social media platform and encounters emotionally charged content.
- **Scenario:**
 1. User highlights the post: *“Breaking news! The government is secretly planning to shut down all banks tomorrow – withdraw your money now before it’s too late!”*
 2. The extension sends the highlighted text to the backend.
 3. The misinformation detection module processes the text, identifying emotional and manipulative cues (“secretly planning,” “too late”).
 4. The model classifies the content as misinformation and flags it as *emotionally manipulative*.
 5. The module returns a JSON response with the classification result and confidence score.

6. A warning icon appears. When clicked, the chatbot opens and provides context: *“This post contains language designed to incite fear (‘too late’, ‘secretly planning’). No credible sources confirm that banks will shut down tomorrow. This is a common misinformation tactic.”*
- **Postcondition:** The user learns that the post uses manipulative language and is unlikely to be factual, encouraging critical evaluation before acting or sharing.

8.5 Test Cases

The following test cases represent a sample of the scenarios that will be used to validate the module's functional requirements.

Table 8.3: Tentative Test Cases for System Validation

Test Case ID	Requirement	Test Description	Expected Result
TC-MIS01	FR2	Input text containing a clear misinformation claim: <i>“Drinking garlic water cures COVID-19 instantly.”</i>	The system classifies the statement as misinformation and highlights manipulative phrase “cures instantly.”
TC-MIS02	FR3	Input text containing emotionally manipulative language: <i>“The government is secretly planning to shut down all banks tomorrow – withdraw your money now!”</i>	The system detects emotionally manipulative terms (“secretly planning,” “too late”) and flags the text as misinformation.
TC-MIS03	FR4	Input text cross-referenced with misinformation datasets: <i>“Vaccines contain microchips to track people.”</i>	The system cross-references known misinformation dataset, confirms the claim as misinformation, and provides confidence score.
TC-MIS04	FR2, FR3	Input text with neutral information: <i>“The weather in Colombo is warm today.”</i>	The system returns no misinformation detected. (Negative test case).

TC-MIS05	FR5	Input text flagged as misinformation.	The system outputs structured JSON with classification result, manipulative features, and confidence score
TC-MIS06	FR6, FR7	User highlights a suspicious claim.	A warning icon appears; clicking it opens the chatbot interface which explains why the text may be misinformation.

8.6 Wireframes

The user interface for the misinformation detection alerts is designed to be minimal, intuitive, and non-intrusive. The wireframes in **Appendix B** illustrate the intended design, showing:

- The appearance of a subtle **warning icon** next to the user’s highlighted text when potential misinformation is detected.
- The interactive **chat-based explanation window**, which opens upon clicking the icon, provides the user with a clear, contextualized explanation of why the highlighted text may be misinformation.

This design emphasizes usability and user education, ensuring that the detection system acts as a supportive guide rather than a disruptive barrier.

9. PROJECT PLAN AND WORK BREAKDOWN STRUCTURE (WBS)

This section outlines the detailed plan for executing the project, from initial research to final delivery. The project is divided into five distinct phases. The following Work Breakdown Structure (WBS) details the key tasks within each phase, assigns responsibilities to team members, and provides estimated durations. A detailed timeline is visualized in the Gantt Chart located in **Appendix A**.

(Note: M1 = Herath (PII/OpSec, Lead), M2 = Member 2 (Hate Speech), M3 = Member 3 (Misinformation), M4 = Member 4 (Scam, UI))

Phase 1: Research and Foundation (Duration: 4 Weeks)

Table 9.1: Phase 1 of the Work Breakdown Structure: Research and Foundation

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
1.1	Comprehensive Literature Survey for all modules	All	Finalized Reference List, Annotated Bibliography	2 Weeks
1.2	Technology Stack Setup & Environment Configuration	M1	Shared GitHub repository, Configured dev environments	1 Week
1.3	Finalize System Architecture and API "Contract"	M1	Final System Diagrams, API Specification Document	1 Week
1.4	Initial Data Sourcing and Feasibility Assessment	All	Report on available datasets for each module	1 Week

Phase 2: Core Module Development (Duration: 6 Weeks - Parallel Work)

Table 9.2: Phase 2 of the Work Breakdown Structure: Core Module Development

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
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2.1	PII/OpSec: Data Synthesis & Annotation	M1	Annotated training and test dataset (JSONL format)	2 Weeks
2.2	PII/OpSec: Model Development & Implementation	M1	Python script for the hybrid NER/Regex model	3 Weeks
2.3	PII/OpSec: Rule Engine Implementation	M1	Python script for the OpSec contextual rule engine	1 Week
2.4	Hate Speech: Model Fine-tuning and Training	M2	Trained hate speech classification model	6 Weeks
2.5	Misinformation: Model Fine-tuning and Training	M3	Trained manipulative language model	6 Weeks
2.6	Scam: Model Fine-tuning and Training	M4	Trained scam detection model	6 Weeks

Phase 3: System Integration and Frontend Development (Duration: 4 Weeks)

Table 9.3: Phase 3 of the Work Breakdown Structure: System Integration and Frontend Development

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
3.1	Backend API Development (FastAPI)	M1, M3	Functional API endpoints for all modules	2 Weeks
3.2	Browser Extension Frontend Development (JS/HTML/CSS)	M4, M2	UI shell of the browser extension	2 Weeks
3.3	Integrate All Modules with the Backend API	M1	Integrated backend service	1 Week

3.4	Connect Browser Extension to Backend API	M4	End-to-end data flow from browser to backend	1 Week
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Phase 4: Testing and Evaluation (Duration: 3 Weeks)

Table 9.4: Phase 4 of the Work Breakdown Structure: Testing and Evaluation

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
4.1	PII/OpSec: Unit & Integration Testing	M1	Unit test report, Successful integration test logs	1 Week
4.2	PII/OpSec: Performance Evaluation & Benchmarking	M1	Evaluation report with Precision, Recall, F1Score	1 Week
4.3	End-to-End System Testing	All	Bug report, System performance metrics	2 Weeks
4.4	Qualitative User Feedback Collection & Analysis	M4	Survey results summary	1 Week

Phase 5: Finalization and Reporting (Duration: 4 Weeks)

Table 9.5: Phase 5 of the Work Breakdown Structure: Finalization and Reporting

Task ID	Task Description	Lead	Deliverable(s)	Estimated Duration
5.1	Final Project Report Writing	All	Draft report sections	2 Weeks
5.2	Final Report Compilation and Editing	M1	Complete Final Report document	1 Week
5.3	Final Presentation and Viva Preparation	All	Presentation slides, Demonstration script	2 Weeks
5.4	Final Code Submission and Documentation	All	Well-documented codebase on GitHub	1 Week

10. BUDGET AND BUDGET JUSTIFICATION

This project is designed to be completed with no direct financial cost. The development and execution of the misinformation detection module will rely exclusively on open-source software, publicly available academic datasets, and existing university or personal computing infrastructure.

- **Software Costs:** All required software, including the Python programming language, the FastAPI framework, the Hugging Face Transformers library, and the PyTorch/TensorFlow deep learning frameworks, are freely available under open-source licenses. No commercial software purchases are necessary.
- **Hardware Costs:** Development, fine-tuning, and testing will be performed using personal laptops and standard university-provided computing facilities. The computational requirements for the prototype stage can be met without specialized or high-performance hardware.
- **Data Acquisition Costs:** Training and evaluation datasets for misinformation detection will be sourced from publicly available research repositories (e.g., LIAR, CoAID, FakeNewsNet, and Kaggle) along with the creation of a synthetic dataset tailored to the project's objectives. These datasets are openly accessible and incur no acquisition cost.

Therefore, the proposed budget for this research project is **zero**, as all essential resources can be obtained at no financial expense.

11. COMMERCIALIZATION AND ENTREPRENEURSHIP POTENTIAL

The **Misinformation Detection Module** within The Guardian system holds strong commercialization potential as a proactive, educational, and user-focused defense mechanism that addresses the global problem of online misinformation. Unlike existing fact-checking platforms that act reactively, this module delivers real-time, personalized, and interactive feedback, positioning it as a transformative tool in the growing digital literacy and information security markets.

11.1 Market Opportunity

With global social media users projected to approach **six billion by 2027** [14], the scale of misinformation exposure is unprecedented. Surveys indicate that over **60% of internet users encounter misinformation weekly**, with health, politics, and finance being the most affected domains [15]. Current solutions—such as third-party fact-checking websites—are often slow, reactive, and inaccessible to everyday users. The *Guardian*'s misinformation module directly fills this gap by providing **real-time, user-facing, and educational explanations**, creating a major opportunity within the intersection of cybersecurity, digital literacy, and consumer protection.

11.2 Commercialization Pathways

The module's commercialization strategy leverages established subscription-based cybersecurity business models while also addressing the rising demand for specific **media literacy tools**.

- **Freemium Browser Extension (B2C):**
 - **Free Tier:** Allows users to highlight and analyze suspicious text, providing core misinformation detection and basic chatbot explanations.
 - **Premium Tier:** Subscription (\$2–5/month) unlocks advanced features, including deeper linguistic analysis, access to

misinformation trend dashboards, and tailored critical-thinking guidance.

- **Enterprise & Institutional Licensing (B2B):**
 - **Corporate Security Training:** License the module to organizations as part of employee training to mitigate risks from misinformation-driven phishing or manipulation.
 - **Educational Integration:** Partner with universities, schools, and digital literacy initiatives to integrate the module into curricula, fostering critical thinking and safe media consumption.
- **API & Module Licensing (Technology Partnerships):**
 - **Cybersecurity Firms:** License the specialized detection modules as APIs to established security companies to enhance their existing product suites.

11.3 Value Proposition

The misinformation detection module creates value by addressing misinformation as a **human-layer cybersecurity threat**, combining AI-driven analysis with user education.

- **Empowered Users:** Helps individuals critically evaluate suspicious claims instead of passively consuming them.
- **Real-Time Guidance:** Delivers conversational explanations instantly, bridging the gap between fact-checking and digital literacy.
- **Non-Intrusive Experience:** Provides subtle alerts and chat-based feedback without censoring content or limiting user autonomy.
- **Scalability:** The modular design ensures seamless integration with other Guardian components or third-party platforms.

11.4 Entrepreneurial Impact

The misinformation detection module can evolve beyond an academic prototype into a viable cybersecurity and digital literacy product. Its innovative blend of **AI, psychology, and usability principles** makes it attractive for **venture funding, startup incubators, and NGO-backed media literacy programs**. With rising global concern about election manipulation, health misinformation, and fake news, the module addresses a **critical and expanding market need**, positioning *The Guardian* as a pioneering solution in the human-layer security landscape.

REFERENCES

- [1] S. van der Linden, A. Leiserowitz, E. Rosenthal, and A. Maibach, “Long-term effectiveness of inoculation against misinformation: Three longitudinal experiments,” *Political Psychology*, vol. 41, no. S1, pp. 17–37, 2020.
- [2] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter,” *arXiv preprint arXiv:1910.01108*, 2019.
- [3] N. Ruchansky, S. Seo, and Y. Liu, “CSI: A hybrid deep model for fake news detection,” in *Proc. 26th ACM Int. Conf. on Information and Knowledge Management (CIKM)*, 2017, pp. 797–806.
- [4] W. Y. Wang, “Liar, liar pants on fire: A new benchmark dataset for fake news detection,” in *Proc. 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2017, pp. 422–426.
- [5] K. Shu, D. Mahudeswaran, S. Wang, D. Lee, and H. Liu, “FakeNewsNet: A data repository with news content, social context, and dynamic information for studying fake news on social media,” *Big Data*, vol. 8, no. 3, pp. 171–188, 2020.
- [6] J. Cui and B. Lee, “CoAID: COVID-19 healthcare misinformation dataset,” *arXiv preprint arXiv:2006.00885*, 2020.
- [7] M. Zeng, K. Starbird, and E. Spiro, “#Unconfirmed: Classifying rumor stance in crisis-related social media information,” in *Proc. 10th Int. AAAI Conf. on Web and Social Media (ICWSM)*, 2016, pp. 747–750.
- [8] A. Rashkin, E. Choi, J. Y. Jang, S. Volkova, and Y. Choi, “Truth of varying shades: Analyzing language in fake news and political fact-checking,” in *Proc. Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, 2017, pp. 2931–2937.
- [9] T. Mitra and E. Gilbert, “CREDBANK: A large-scale social media corpus with associated credibility annotations,” in *Proc. 9th Int. AAAI Conf. on Web and Social Media (ICWSM)*, 2015, pp. 258–267.
- [10] S. Vosoughi, D. Roy, and S. Aral, “The spread of true and false news online,” *Science*, vol. 359, no. 6380, pp. 1146–1151, 2018.
- [11] J. Thorne, A. Vlachos, C. Christodoulopoulos, and A. Mittal, “FEVER: A large-scale dataset for fact extraction and verification,” in *Proc. 16th Conf. of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2018, pp. 809–819.
- [12] S. Kumar and N. Shah, “False information on web and social media: A survey,” *Proc. VLDB Endowment*, vol. 10, no. 12, pp. 1716–1727, 2017.

- [13] Z. Zhou, J. Wu, and W. Zafarani, “SAFE: Similarity-aware multi-modal fake news detection,” in *Proc. 24th Pacific-Asia Conf. on Knowledge Discovery and Data Mining (PAKDD)*, 2020, pp. 354–367.
- [14] Statista, “Number of social media users worldwide from 2017 to 2027,” *Statista Research Department*, 2023. [Online]. Available: <https://www.statista.com/statistics/278414/number-of-worldwide-social-network-users/>
- [15] PwC, “Global digital trust insights survey 2023,” *PricewaterhouseCoopers*, 2023. [Online]. Available: <https://www.pwc.com/digitaltrustinsights>
- [16] McKinsey & Company, “Cybersecurity trends: Looking over the horizon,” *McKinsey Insights*, 2022. [Online]. Available: <https://www.mckinsey.com/business-functions/risk-and-resilience>
- [17] Verizon, “Data Breach Investigations Report (DBIR) 2022,” *Verizon Enterprise*, 2022. [Online]. Available: <https://www.verizon.com/business/resources/reports/dbir/>
- [18] A. Beaument, M. A. Sasse, and M. Wonham, “The compliance budget: Managing security behavior in organizations,” in *Proc. 2008 Workshop on New Security Paradigms (NSPW)*, 2009, pp. 47–58.

Appendix A: Project Gantt Chart

Group ID: 25-26J-041

* Owner key:

Members: M1 = Group Group/ Module-A lead -Herath H M G D K (IT22361486),
M2 = Module-B lead - De Alwis H P G H S (IT2237926),
M3 = Module-C lead - Fernando M R I (IT22366054),
M4 = Module-D = UI/UX lead - Vishwa J R (IT22361612)

Project Start:	Mon, 16/06/2025
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Display Week: 1

Insert new rows ABOVE this one

Appendix B: UI Wireframes



Fig. B.1. Wireframe of the initial states that user highlights the text that the user would like to verify.

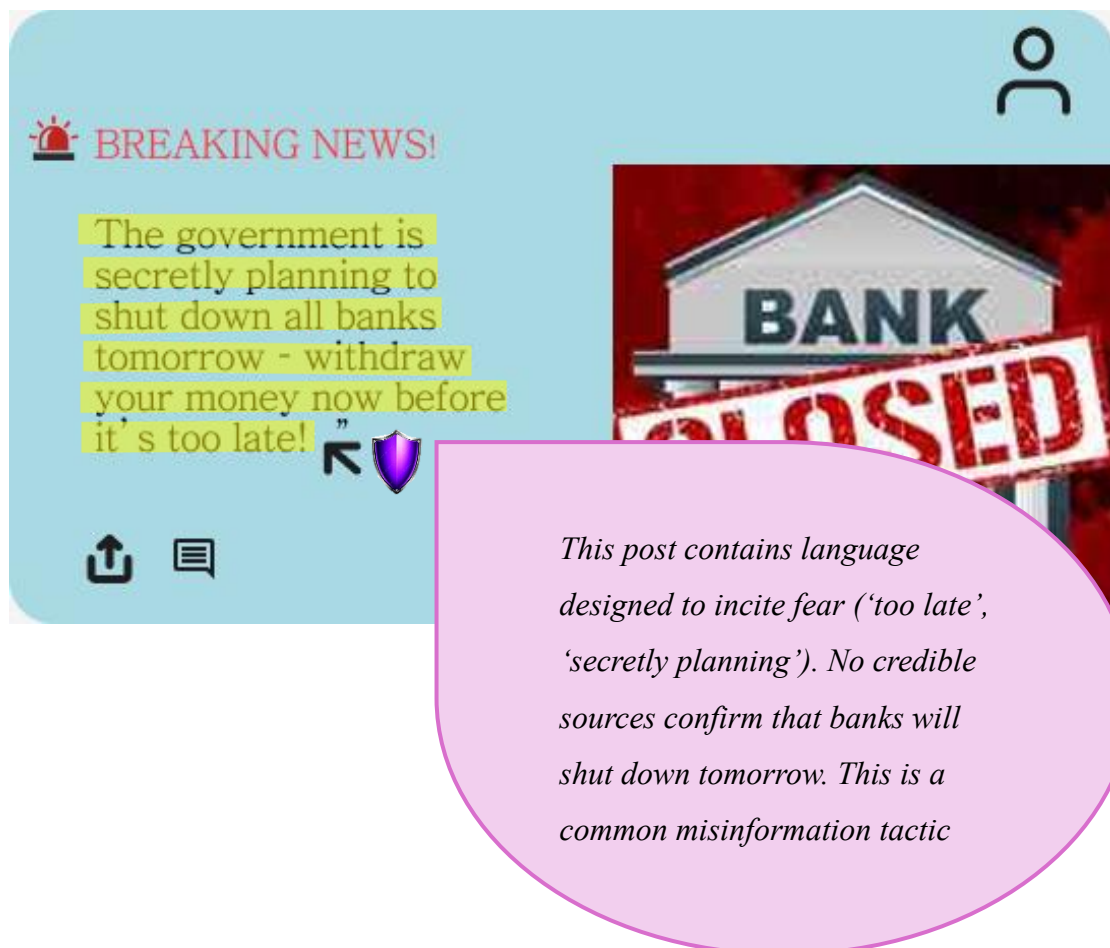


Fig. B.2. Wireframe of the hover state. When the user interacts with the shield icon, a chat opens and provides a concise, educational explanation of the specific risk according to the user personalization, empowering the user to make an informed decision.